

TATA CONSULTANCY SERVICES

EXECUTIVE REPORT

TCS ACTIVE SOVEREIGNTY · HIGH-TECH & SEMICONDUCTORS

The Minimum Viable

# Semiconductor Equipment Maker

What must never stop — **tools built, shipped and serviced worldwide** — and how to keep it running under your own control.

01 EXECUTIVE SUMMARY

# Sovereignty is now a board decision

Every organisation has a short list of things that must never stop. For a semiconductor equipment maker, it starts with **tools built, shipped and serviced worldwide**. Active Sovereignty keeps that list running — under your own control — whatever happens to suppliers, jurisdictions or the threat landscape.

Sovereignty used to be a compliance question about where data is stored. It is now a strategic question of continuity and control: who could switch off, read or change the systems the business depends on — and how fast it could recover if they did. For semiconductor and high-tech companies, the European Chips Act, export-control regimes and the Cyber Resilience Act make control over design IP, installed-base data and supply chains a strategic and geopolitical question.

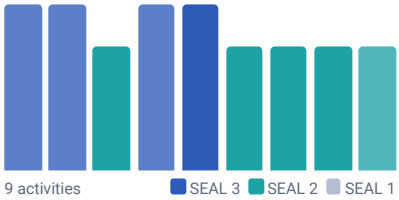
The answer is not to move everything into a sovereign environment. It is to define the **Minimum Viable Enterprise (MVE)**: the smallest set of capabilities, data and operations that must stay available and under European control. For a semiconductor equipment maker, **4 of 9** activities form that load-bearing core, led by export-controlled design & engineering IP, installed-base remote service & machine data and production & system integration — including the often-overlooked

One size does not fit all. **5 of 9** activities stay on public or public sovereign cloud; only the core moves to private or national. No activity is targeted above SEAL 3 — chips, hardware and much software and SaaS remain non-EU.

TCS recommends what is right for each activity; the board decides what to implement, what to phase and which risks to accept. The cost is driven by how the model is operated far more than by infrastructure — and it can be contained.

### AT A GLANCE

| | |
|---|---|
| 9 | value-chain activities, including the enterprise backbone |
| 4 | load-bearing — target band reaching SEAL 3 |
| 5 | on public and public sovereign cloud |
| 90 | days to an executive decision that sets the direction |




### THREE MESSAGES FOR THE BOARD

1

**Define what must never stop.**

Agree the MVE and its recovery objectives — the commitments you can keep on your worst day.

2

**Protect the core, not everything.**

SEAL 3 only where criticality demands it; the rest on public cloud with proportionate controls.

3

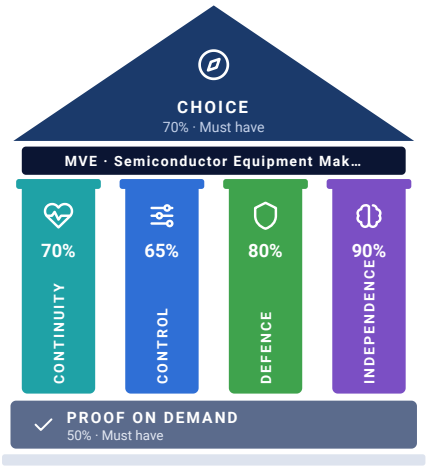
**Make it provable.**

Treat exit, recovery and control as capabilities you evidence continuously — not clauses in a contract.

02 The six questions
03 The path to your MVE
04 Risk posture
05 Resilience & operating model
06 The business case
07 What TCS brings
08 The roadmap
09 Design MVE · 90 days

02 THE MVE AND ITS SIX QUESTIONS

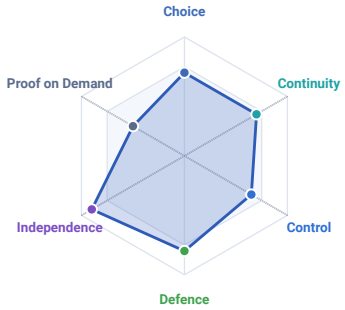
# Six questions only the board can answer




At the centre sits the **Minimum Viable Enterprise** — for a semiconductor equipment maker, **tools built, shipped and serviced worldwide** — held up by six pillars. Each is a question that sits with a different member of the executive team; no single function can answer it alone.

The percentages are the **TCS recommended focus for High-Tech & Semiconductors**: how much weight each question deserves on a scale from nice to have to critical. The board can adjust them; the minimum answers below cannot be left open.

**FIGURE 1** The Enterprise House and the TCS recommended focus




Sovereignty fingerprint · High-Tech & Semiconductors

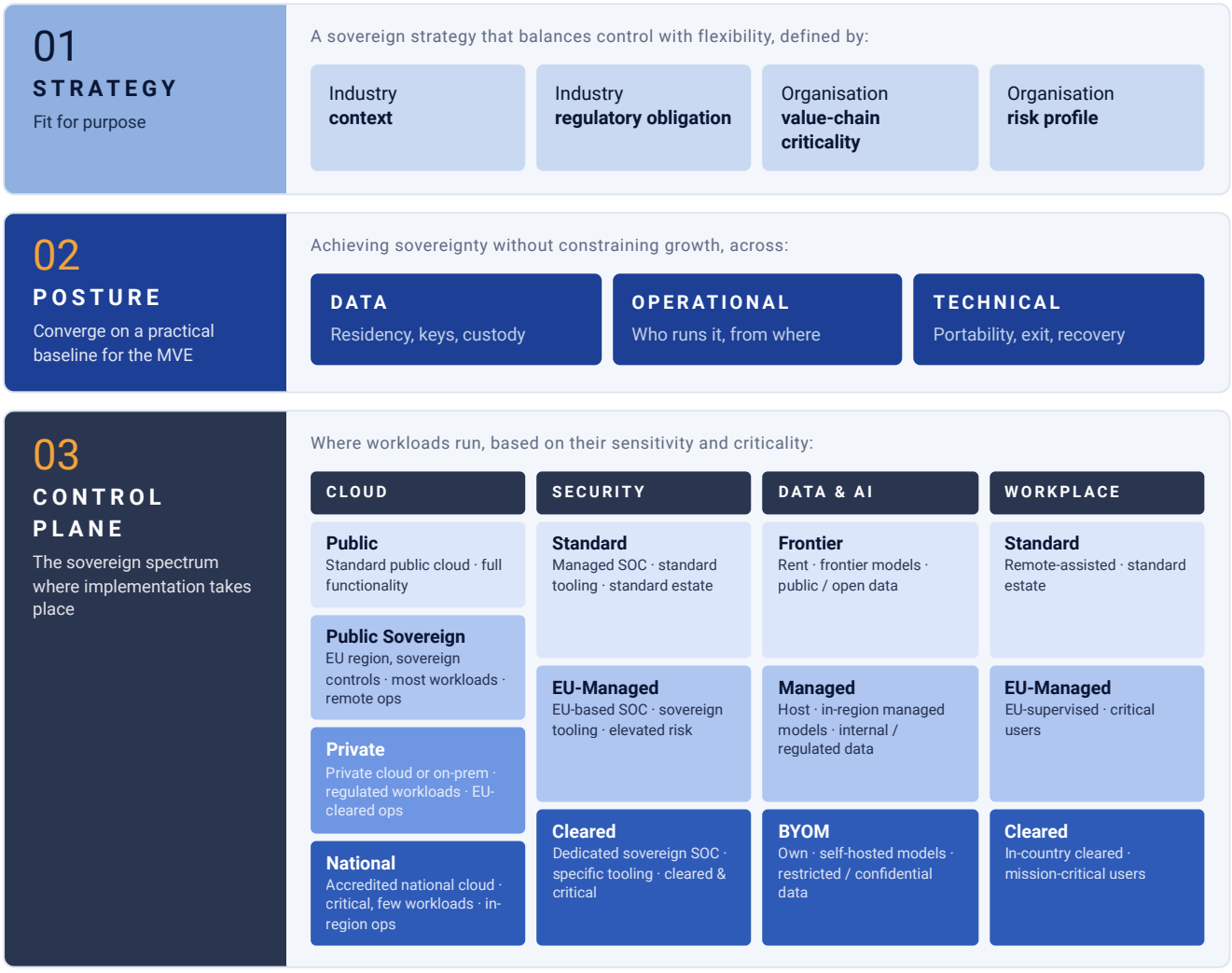
| THE QUESTION | THE MINIMUM ANSWER FOR A SEMICONDUCTOR EQUIPMENT MAKER | TCS FOCUS |
|---|---|---|
| **Whoever changes the rules, can we still choose our own path?** CHOICE · CEO / BOARD | Export-controlled design IP and installed-base data under EU jurisdiction, with no foreign access path. | 70% MUST HAVE |
| **Will the business keep running through shocks we don't control?** CONTINUITY · COO | Customer fabs keep running: remote service and spares continue through a provider or cyber disruption. | 70% MUST HAVE |
| **Do we command the technology we run on — or does our provider?** CONTROL · CIO / CTO | Engineering and service platforms run by EU-cleared operators; strict export-control segregation. | 65% MUST HAVE |
| **If we're attacked or cut off, can we defend and restore ourselves?** DEFENCE · CISO | Isolated recovery of PLM, design and service systems; zero-trust access for partners. | 80% CRITICAL |
| **Can we harness the best data and AI without becoming hostage to it?** INDEPENDENCE · CDO / CAIO | Machine data, design IP and AI models owned and governed by the company. | 90% CRITICAL |
| **Can we prove sovereignty actively, on demand — not once a year?** PROOF ON DEMAND · CRO / CCO | Evidence for export authorities, customers and governments on demand. | 50% MUST HAVE |

03 TCS ACTIVE SOVEREIGNTY FRAMEWORK

# The path to your Minimum Viable Enterprise

Active Sovereignty is sovereignty that is not only designed, but **continuously operated, controlled and demonstrably assured**. The path runs through three layers — from strategic intent, to a practical posture, to the control plane where it is implemented.

**FIGURE 2** The path to MVE for Active Sovereignty




**From measured exposure to enforceable control**

### 01 Strategy — fit for purpose

Sovereignty is a continuum, not a switch. The right level depends on the industry, its regulation, how critical each part of the value chain is and the organisation's risk profile. For semiconductor and high-tech companies, the European Chips Act, export-control regimes and the Cyber Resilience Act make control over design IP, installed-base data and supply chains a strategic and geopolitical question.

### 02 Posture — a practical baseline

Every activity gets a target across three dimensions: **Data** — where it resides and who holds the keys; **Operations** — who runs it and from where; **Technology** — how it is recovered, moved or exited.

### 03 Control plane — where it becomes real

Each workload is placed across four planes. Most run on public or public sovereign cloud with sovereign controls; the control-critical core on private cloud or on-prem; national cloud only for strict national-control workloads.

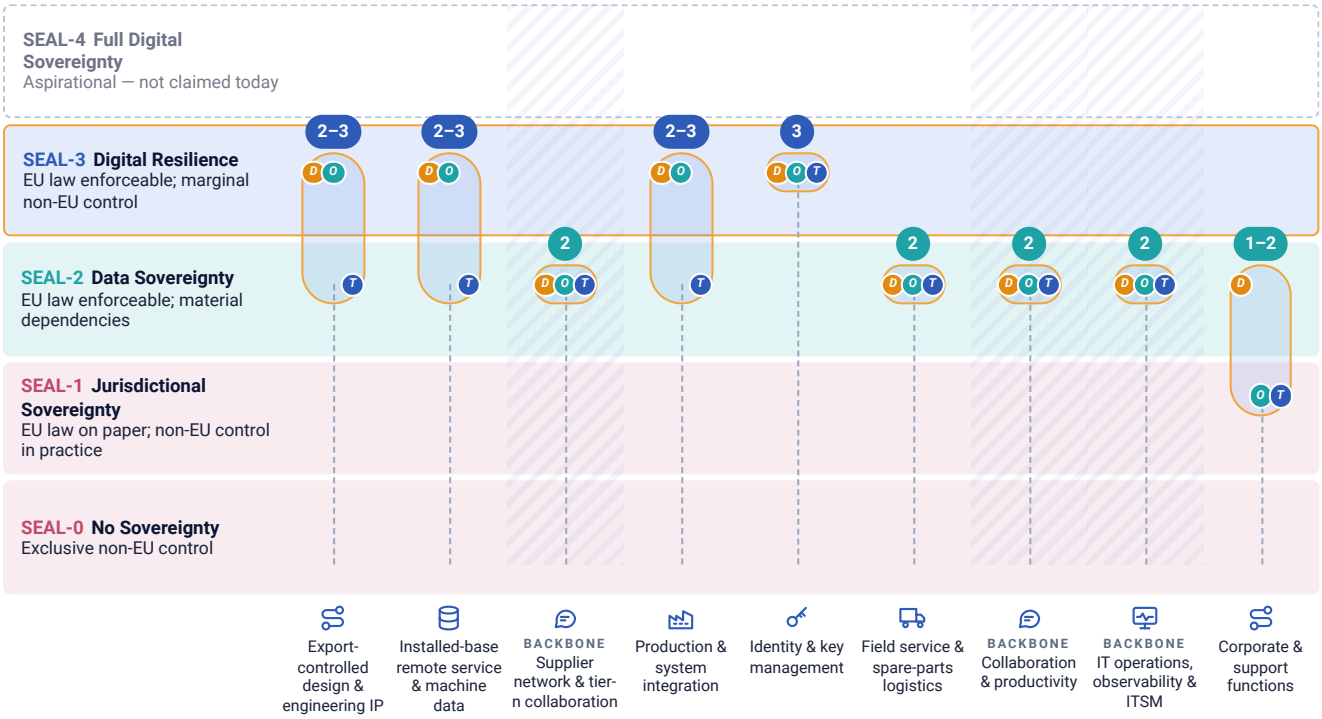
04 RISK POSTURE

# Protection matched to criticality

SEAL-4 needs no critical non-EU dependency in chips, hardware, software or SaaS, and is not attainable today — so **no activity is targeted above SEAL 3**. Each activity below shows its Data, Operations and Technology targets as whole levels; the band spans them.

**Recommendation based on the TCS Active Sovereignty Framework** · High-Tech & Semiconductors

**FIGURE 3** Target SEAL per activity — industry core plus the enterprise backbone (shaded)




| ACTIVITY | TARGET | D · O · T | WHERE IT RUNS | WHY THIS TARGET |
|---|---|---|---|---|
| **Identity & key management** | SEAL 3 | 3 · 3 · 3 | Private | Root of trust. |
| **Export-controlled design & engineering IP** | SEAL 2–3 | 3 · 3 · 2 | Private | Geopolitically strategic IP. |
| **Installed-base remote service & machine data** | SEAL 2–3 | 3 · 3 · 2 | Private | Customer fabs depend on it. |
| **Production & system integration** | SEAL 2–3 | 3 · 3 · 2 | Private | Every system is unique and high value. |
| **Supplier network & tier-n collaboration** ENTERPRISE BACKBONE | SEAL 2 | 2 · 2 · 2 | Public Sovereign | Thousands of specialised suppliers. |
| **Field service & spare-parts logistics** | SEAL 2 | 2 · 2 · 2 | Public Sovereign | Uptime is the customer promise. |
| **Collaboration & productivity** ENTERPRISE BACKBONE | SEAL 2 | 2 · 2 · 2 | Public Sovereign | Mail, files, chat and copilots (e.g. Microsoft 365, Teams, SharePoint, Copilot, Google Workspace) hold board papers… |
| **IT operations, observability & ITSM** ENTERPRISE BACKBONE | SEAL 2 | 2 · 2 · 2 | Public Sovereign | Monitoring, logging, APM, CMDB and service-management tools hold the map of the estate: topology, credentials in… |
| **Corporate & support functions** | SEAL 1–2 | 2 · 1 · 1 | Public | Important, not load-bearing. |

05 RESILIENCE & OPERATING MODEL

# What holds under stress — and where it runs

A posture is only as good as its behaviour under stress. The MVE keeps the core running through each disruption below; where it adds real value, TCS recommends a **fallback on top**. We recommend — the board decides.

**FIGURE 4** The semiconductor equipment maker MVE as a deliberate mix — number of activities per tier, darker = more of the MVE




**FIGURE 5** Disruption scenarios — what the recommended MVE delivers, and where a fallback is recommended

| DISRUPTION | WITH THE RECOMMENDED MVE | RECOMMENDED FALLBACK |
|---|---|---|
| **Cloud provider cut off** *Without an MVE: Core services stop until the provider relationship is restored.* | The core already runs on EU-operated private and national environments; channels on public sovereign cloud fail over within agreed timeframes. | Continuity pod — a minimal standby of the core on a second EU-operated environment. |
| **Ransomware hits the core** *Without an MVE: Core systems and backups are encrypted; recovery time is unknown.* | An immutable vault and clean-room rebuild restore the core in agreed priority order. | Break-glass identity held outside the primary platform, so access can be restored first. |
| **Foreign access to data or keys** *Without an MVE: Data, documents, operational metadata and keys can be compelled outside EU law.* | Hold-your-own keys and EU-cleared operations remove the access path. | No fallback needed — the MVE holds. |
| **AI model withdrawn** *Without an MVE: AI-dependent decisions and copilots stop or degrade without warning.* | Critical AI workloads run on managed or owned models with portable data and prompts. | A privately deployed model (BYOM) as a switch-over target for the most critical AI use cases. |
| **Critical SaaS provider exits** *Without an MVE: Core processes, documents and their data are locked in an unavailable platform.* | Data is held in portable formats under your control, with a tested exit plan for each critical SaaS provider. | An in-country standby instance with daily data export for the processes that cannot stop. |
| **Regulator asks for proof — today** *Without an MVE: Evidence is assembled manually, under deadline pressure.* | Continuous evidence on control, exit and recovery is available on demand. | No fallback needed — the MVE holds. |

06 THE BUSINESS CASE

# Contained cost, not blanket cost

Boards rightly ask what sovereignty costs. The cost is driven far less by infrastructure than by the **operating model around it** — and most of it recurs every year. The decisive question is how to contain the recurring cost of running the MVE.

**FIGURE 6** What drives the cost of a sovereign operating model — and how TCS contains it

### What drives the cost up

| | |
|---|---|
| **Running & protecting the estate**<br>Local, security-cleared teams around the clock | *Ongoing — the largest line* |
| **Identity & access control**<br>A separate, customer-controlled sign-in and key system | *Set-up + ongoing* |
| **Security & isolation**<br>Dedicated protection for the most sensitive systems | *Set-up + ongoing* |
| **Business continuity & recovery**<br>Independent, regularly tested recovery | *Set-up + ongoing* |
| **Audit & compliance**<br>Evidence and independent checks | *Ongoing — grows as rules tighten* |
| **Keeping data & keys in-region**<br>In-region data, backups and replication | *Set-up + ongoing* |
| **Platform & infrastructure**<br>Dedicated capacity or more controlled ground | *Set-up + ongoing* |
| **Keeping sovereignty current**<br>Re-certification and rehearsals | *Ongoing* |
| **Software & suppliers**<br>Sovereign editions and vendor assurance | *Ongoing — only where needed* |

### How TCS contains it

**TCS SOVEREIGNSECURE CLOUD™**

- **Build-Public, Run-Sovereign** — build on flexible cloud; run hardened workloads sovereign
- **Shared Sovereign Platform** — cleared teams and infrastructure shared across organisations
- **Open Technology** — open-source core; sovereign editions only where required

**SOVEREIGN OPERATIONS · TCS EDEN™**

- **Autonomous Operations** — AI-led operations; cleared teams handle exceptions
- **AI Service Experience** — AI-first service desk
- **Always-On Resilience** — recovery continuously and automatically proven

**SOVEREIGN SECURITY**

- **Autonomous Security** — threats detected and contained automatically
- **Unified Sovereign Identity** — one identity and key platform

**AUDIT & COMPLIANCE**

- **Policy-as-Code** — evidence generated by the platform
- **Continuous Assurance** — automated drift detection

**FIGURE 7** What a credible sovereignty business case shows

1
**The cost of not acting**
Outage, exit and supervisory exposure if the core fails or is cut off.

2
**The overlay — for the MVE only**
Sovereign controls on the load-bearing core, not the whole estate.

3
**The value created**
Continuity, market access, supervisor confidence and freedom of choice.

4
**The funding gates**
Investment released in waves, as risk and value demand.

**WE CONTAIN THE COST; WE DO NOT CLAIM TO REMOVE IT**

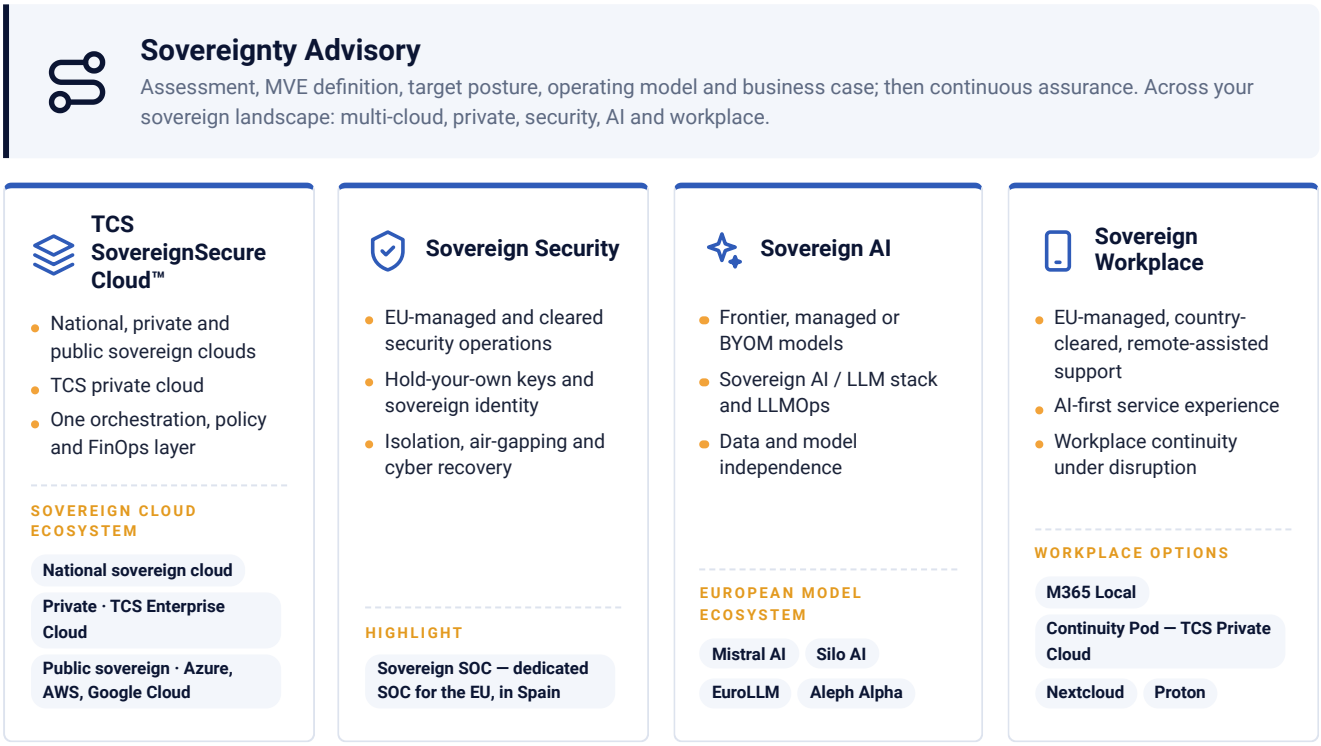
The first lever is the MVE itself: **protect the core, not everything**. Over five years the recurring cost of sovereign operations rises with the cost of living; as AI and automation mature, efficiency compounds and bends it down.

07 WHAT TCS BRINGS

# Sovereignty where it matters, flexibility where it helps

Advisory, four implementation planes and sovereign operations under one governed model — **from executive intent to a running Active Sovereign MVE**, on a strong European operations backbone.

**FIGURE 8** The TCS Active Sovereignty value proposition




**Advisory sets the direction.** Every engagement starts with intent: which activities must hold, to what level, at what cost. Sovereignty Advisory turns that into the MVE, the target posture, the operating model and the business case — and later into continuous assurance, so the posture does not drift.

**Four planes deliver it.** TCS SovereignSecure Cloud™ — including TCS Private Cloud alongside national and public sovereign clouds — Sovereign Security with a dedicated SOC for the EU, Sovereign AI and Sovereign Workplace implement the posture, activity by activity, on the European ecosystem.

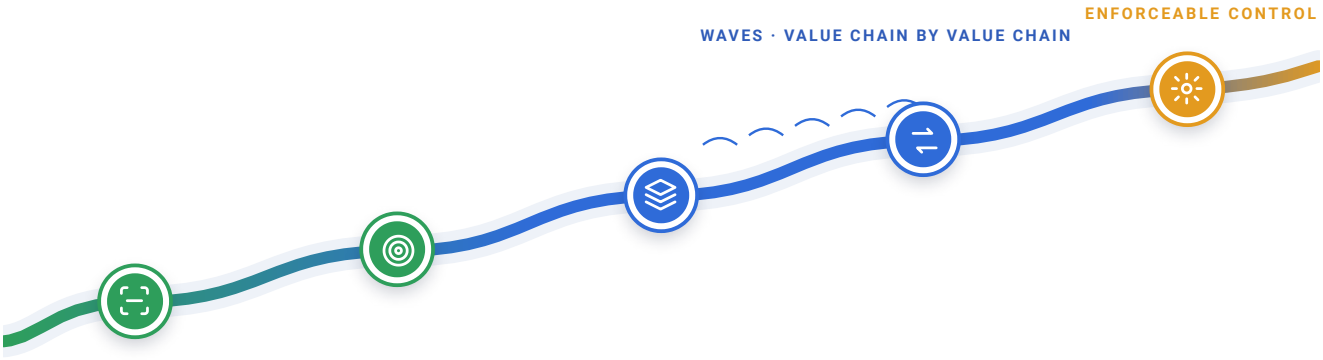
**TCS EDeN™ runs it.** Sovereignty is only as strong as the operations behind it: 31+ locations and 15,000+ EU-based professionals, 10 availability zones on TCS Private Cloud plus operations on the public clouds, and a dedicated SOC in Spain.

08 RUN ACTIVE SOVEREIGNTY WITH TCS

# The roadmap to Active Sovereignty

A journey, not a project: design the MVE for a semiconductor equipment maker, build resilience in waves — value chain by value chain — then operate sovereignty as **an active state, maintained continuously**, with proof on demand.

**FIGURE 9** From measured exposure to enforceable control




**DISCOVER**
Measure exposure and table the starting point

**STRATEGIZE**
Define the MVE and the target placement

**IMPLEMENT**
Integrate the stack under governed controls

**MIGRATE**
Move or sovereign-ise priority workloads, in waves

**OPERATE**
Sovereignty as an active state — maintained continuously

## Design MVE

Discover · Strategize · Months 0–3

OUTCOMES

- Starting point tabled: exposure measured
- MVE defined — enterprise-wide or one value chain
- Workload placement across the four tiers
- Executive decision: direction set

TCS ACCELERATORS

- TCS Active Sovereignty Framework
- Industry baselines & risk postures
- Sovereign Assessment Agents
- Sovereign Operating Model
- Sovereign Use Case Catalogue
- TCS Hybrid Cloud Advisor

## Build Resilience

Implement · Migrate · ≈ Months 3–18 · in waves

Workloads · wave by wave
Keys & identity
Sovereign landing zone

OUTCOMES

- Sovereign landing zones in place
- Keys & identity under EU control
- Priority workloads migrated or sovereign-ised in place
- Exit runbooks tested

TCS ACCELERATORS

- TCS Sovereign Forge
- Agentic AI Infrastructure for Local
- TCS AgentVMove
- TCS SovereignGuard
- FinOps for AI
- TCS Agentic SDLC
- Agentic Migration Factory · MS · AWS · GCP

## Operate Active Sovereignty

Operate · an active state · Continuous

OUTCOMES

- MVE implemented and active
- Posture maintained continuously — proof on demand
- EU SOC monitoring & resilience drills
- Exit readiness verified, drift corrected

TCS ACCELERATORS

- TCS Alpha
- TCS Iris
- TCS Xponence
- TCS Cognix
- TCS Agentic Operations
- MS Sovereign data controls
- AWS Sovereign Cloud Command Center
- GCP Suite

Indicative timeline — the pace is set per value chain in the assessment. Design MVE — the first stage — is where the starting point is tabled. The next page shows its two scopes.

09 DESIGN MVE · THE 90-DAY ACCELERATION

# Two ways in, one outcome: the direction set

Discover and Strategize, powered by the TCS Sovereignty Assessment, turn the reference MVE in this report into your own — scored on your evidence. It runs in one of two scopes; **the deliverables and the executive decision pack change with it**.

**Before day one**
Agentic pre-assessment — readiness, data, regulation, risk-posture snapshot

**Discovery sprint**
Measure exposure and table the starting point

**Strategy sprint**
Define the MVE, the placement and the operating model

**Day 90**
Direction set

SCOPE · ENTERPRISE-WIDE

## Board sovereignty strategy

Across the whole enterprise: place every workload on the right tier and give the board one sovereignty strategy. Execution then runs in waves, value chain by value chain.

DISCOVERY SPRINT

**Measure enterprise exposure**

- Enterprise workload inventory & criticality
- Exposure heatmap across all value chains
- Data residency & key control, estate-wide
- Regulatory & extraterritorial exposure
- Provider concentration & vendor evidence

STRATEGY SPRINT

**Define the enterprise MVE**

- Workload placement across the four tiers
- Enterprise MVE & D·O·T target bands
- Sovereign operating model & guardrails
- Identity, keys, exit & portability principles
- Roadmap waves & investment envelope

**Design sprint** · Optional · in parallel — Design 2–3 pilots across the planes, so the strategy is grounded in your own estate.

EXECUTIVE DECISION PACK · ENTERPRISE-WIDE

### Board sovereignty strategy

- **Sovereignty strategy & risk appetite** — What the board approves, and why now.
- **Workload placement principles** — Which workloads go to which tier.
- **Exposure today vs target** — Enterprise heatmap, current versus target.
- **Investment envelope** — Cost, trade-offs and funding gates.
- **Wave plan** — Which value chains first, what follows.

**Fits when** the board needs one sovereignty strategy for the whole enterprise before execution starts.

SCOPE · ONE VALUE CHAIN

## Executive proof point

On one value chain: go deep end to end and give the executives a proof point — then scale the approach to the whole enterprise.

DISCOVERY SPRINT

**Map one value chain end to end**

- Activities, data flows & dependencies
- Regulatory exposure of the value chain
- Vendor & SaaS evidence for its providers
- Disruption impact, activity by activity
- Baseline D·O·T per activity

STRATEGY SPRINT

**Design the value-chain MVE**

- Target D·O·T band per activity
- Deployment design across the tiers
- Continuity & exit design
- Identity & key design for the chain
- Business case for the value chain

**Proof-point design** · Core · ready to build — Design the proof point in detail — architecture, controls and a test plan — ready to build in the next phase.

EXECUTIVE DECISION PACK · ONE VALUE CHAIN

### Executive proof point

- **Go / no-go on the proof point** — What to build, and why this value chain.
- **Value-chain MVE design** — Target bands and deployment per activity.
- **Value-chain business case** — Cost, benefit and risk accepted.
- **Exposure today vs target** — For this value chain.
- **How to scale** — From one value chain to the enterprise.

**Fits when** executives need a visible proof point on one critical value chain before scaling.

NEXT STEP

### Start with a conversation about your MVE.

Talk to your TCS team — and bring an executive decision on your Minimum Viable Enterprise to the table in 90 days.

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